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Public policy strategies for next-generation vehicle technologies: An overview of leading markets

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ABSTRACT

Governments apply a range of policy instruments to promote the electrification of personal transport using technologies such as Plug-in Hybrid, Battery Electric, and Fuel Cell Electric Vehicles. We introduce a framework for comparing public policies cross-nationally used to support the research, development, and deployment of electric vehicles. We use this framework to identify whether governments are using similar policy instruments to promote vehicle electrification in the period 2006-2016. We also examine the extent to which policy instruments are neutral across technologies, presenting country case studies for China, the United States, Japan, and, within the regulatory framework of the European Union, Germany, and France. We find that governments have largely adopted technology neutral policies and are employing similar policy mixes in promoting vehicle electrification, despite differences in governing institutions between the countries examined. We conclude by suggesting a future research agenda, including identifying plausible explanations for this outcome.

1. Introduction

The transport sector contributes roughly 23 percent (8.8 Gt) of global life-cycle CO2 emissions, with road transport constituting the largest share (6.5 Gt) (Miller and Façanha, 2014). Next-generation vehicles (NGV), defined here as Plug-in Hybrid Electric Vehicles (PHEVs), Battery Electric Vehicles (BEVs) and Fuel Cell Electric Vehicles (FCEVs), are proposed as key technologies for reducing transport-related CO_2 emissions (Gilbert and Perl, 2007; Sims, 2014). BEVs and FCEVs have the additional advantage of avoiding local particulates generated by conventional diesel and gasoline-fueled vehicles.

Internal combustion engine (ICE) vehicles nevertheless dominate transport globally, and there remain barriers to the emergence of alternative technologies. Passenger NGVs are commonly higher in price compared to ICEs in similar vehicle classes and BEVs are subject to concerns regarding vehicle range (Haugneland and Kvisle, 2013), battery safety, performance, and recharging time (Styczynski, 2015; Öko Institut, 2011). BEVs and FCEVs also require refuelling infrastructure providing low-carbon electricity or hydrogen at scale.

Public policy thus has an important role to play in promoting the de-carbonization of the transport sector in general, and passenger cars/light duty road vehicles specifically. In this paper we compare the mix of policies implemented by national governments in leading automotive markets to promote NGVs, with the goal of understanding whether governments are adopting similar mixes of policies in promoting de-carbonization. We do so in recognition of the importance of cross-national comparison as a source of

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learning in public policy (Dolowitz and Marsh, 2000).

We also examine whether policies being implemented by key governments are technology-neutral. Technology neutrality is an important principle, based on the idea that the forces of supply and demand, rather than technology choices made by governments, should be the central mechanism selecting which technologies emerge as dominant in new product markets. The benefits of supporting multiple technologies may nevertheless fall as a product matures (Van der Vooren et al., 2012). Technology neutrality is thus an important policy choice governments face as they weigh the costs and benefits associated with supporting vehicle electrification.

Our findings suggest that national governments in major automotive markets are adopting remarkably similar portfolios of policies in promoting NGVs, despite differences in political and economic institutions. Our findings also suggest governments remain committed to technology neutral approaches to promoting NGVs. Future research could examine the plausibility of explanations for these outcomes, with possibilities including cross-national policy learning, and the globalized industry structure of the automotive sector. Future research could also expand the study to include alternative fuels technologies, policies pursued by sub-national governments, and the effectiveness of policies on market penetration across different technology types.

In the balance of the article we discuss recent findings in the relationship between public policy and NGVs. We then introduce a framework for comparing public policies in NGVs, and discuss qualitatively the policies used by central governments to promote NGVs across major automotive markets. In the final section we summarize our findings, and discuss its implications.

2. Context

Scholars have used single country studies to produce important insights into the policies governments are using to promote the decarbonization of passenger vehicles globally. Norway has achieved a high level of BEV penetration through generous consumer subsidies, for example, although research suggests entrepreneurialism by niche actors was an additional important factor (Figenbaum, 2017). The case of France suggests the national government is playing a central role in promoting vehicle electrification, and that it balances electrification with other public policy goals such as affordability (Schamp, 2014). In China, policies towards BEVs are argued to have lacked coherence (Wan et al., 2015).

Cross-national comparison enables empirical and theoretical generalization from single country studies (Wesseling, 2016). Research on multiple countries suggests that national differences can lead countries to adopt different transition pathways in pursuing vehicle electrification (Altenburg et al., 2015). Differences in technology-types also matter: technologies such as biofuels and hybrid electric vehicles, for example, are found to have inherently different systems of governance, in part because of the different constellations of policy actors involved in their development (Nilsson, 2012).

One important empirical issue taken up in studies of transport electrification is whether countries are employing similar portfolios of policy instruments. This matters because differences in the timing and types of policies employed could plausibly affect the development of domestic industries, and the pace of NGV deployment, with implications for national competitiveness and de-carbonization. Early studies suggest that governments have employed different policy mixes. Hildermeier and Villareal (2011), for example, find the French government focused more on demand creation relative to Germany, which concentrated on subsidizing product innovation. In a sample of European countries, Van den Steen et. al. (2015) found governments apply different policy mixes in promoting NGVs that are not necessarily internally coherent, although they have a common goal of overcoming barriers to consumer adoption. Wesseling (2016) examines the 2008–2014 period, similarly finding differences across countries, with some countries focused on technology development, and others on demand creation.

These results contrast with Leurent and Windisch (2011) who conclude governments use a similar basket of policies in promoting PHEVs and BEVs, focused on demand and supply-side measures combined with support for the investments in charging infrastructure. Mock and Yang (2014) also find governments use a common mix of policies, although they focus solely on fiscal incentives, and suggest there are differences in levels of support across different policy types. In countries lacking a domestic automotive sector, governments have a common focus on infrastructure development and the provision of consumer incentives, while unsurprisingly lack policies aimed at industrial innovation (Begley and Donnelly, 2011).

These different results may be attributable to the different range of countries included across studies. A second reason may be found in the different strategies used to compare policies across countries, with some studies comparing policies informally or focusing on a subset of policies, with others adopting more formal and comprehensive approaches to categorizing policy types. Finally, the difference may lie in the periods compared, with some studies focused on earlier periods of NGV development and deployment, and others examining policies in more recent years.

Another important question, in addition to whether governments are using similar policies, to promote NGVs, is whether policies are consistent with the principle of technological neutrality. Technology neutrality is important because technology trajectories can be contingent on which products gain an initial advantage in market share. This can lead to technology lock-in, which is risky in industries that exhibit network externalities such as the automotive sector. The need for vehicle charging infrastructure, in addition to increasing economies of scale, learning effects, and other factors illustrate this risk in the case of vehicle electrification (Klitkou et al. 2015). Technology lock-in can also produce sub-optimal environmental outcomes (Sandén and Azar, 2005; Fox et al., 2017). Among the reasons why ICEs are dominant in passenger transport despite lower environmental performance are factors such as scale economics and the well-developed cross-nationally compatible charging infrastructure.

The neutral treatment in policy of like-products with different technologies during the early stage of market development helps to avoid the risk of sub-optimal technology lock-in (Van der Vooren et al., 2012). The costs of retaining diversity in technology types increase, however, as economies of scale increase. Consumer uncertainty may also slow adoption rates in the presence of different product architectures. The benefits of maintaining technology neutrality may thus fall over time (Van der Vooren et. al., 2012).

Table 1
Descriptive Data for Country Cases (Major Vehicle Markets).
Source: CAAM (2018a,b), USDOT (2017), AFDC (2017); JAMA (2018); NEV (2018); KBA (2018); CCFA (2018); SOeS (2018).

Country	No. of Passenger Car Sale	No. of PHEV Sales	No. of BEV Sales	No. of FCEV Sales	Major Auto Companies
China	24.7 mill. (2017)	111,000 (2017)	468,000 (2017)	N.S.	BAIC, Beijing Automotive Group, Brilliance Automotive, BYD, Chang'an, Chery, Denza, Dongfeng Motor, FAW Group, Geely (Zhidou Electric Vehicle Co.), Great Wall, Guangzhou Automobile Group, Hawtai, JAC, Jianghuai (JAC), Kandi Technologies Group, Lifan, SAIC Motors, Zotye
United States	7.5 mill. (2015)	42,825 (2015)	71,044 (2015)	90 (2015)	Chrysler, Ford, General Motors, Tesla
Japan	4.1 mill. (2016)	13,847 (2016)	13,056 (2016)	1.204 (2016)	Toyota Group, Mitsubishi, Mazda, Nissan, Honda, Suzuki
Germany	3.4 mill. (2017)	25,056 (2017)	29,436 (2017)	N.S.	VW Group, Mercedes-Benz, BMW
France	2.1 mill (2017)	11,817 (2017)	24,910 (2017)	N.S.	PSA Peugeot-Citroën, Renault

Comparing the extent to which national policies are promoting technology neutrality offers a method for identifying how national governments are weighing up the benefits and costs of retaining diversity in technology choices in NGVs.

3. Framework and methods

In this paper we identify and categorize national policies adopted in major vehicle markets to promote research and development into, and the adoption of NGVs. Major vehicle markets are defined as those hosting the headquarters of major automotive production companies, and those with large potential and leading consumer markets, emphasizing the gist of lead market definitions discussed in Quitzow et al. (2014). We thus select China, the United States, Japan, Germany, and France (Table 1). We focus on these markets because they are empirically important: choices made by governments, producers, and consumers in these countries will have a larger effect on aggregate levels of greenhouse gas (GHG) emissions globally from the transport sector, compared to smaller markets, other things being equal.

In order to compare policies, it is necessary to develop analytical categories that can be applied cross-nationally. In general terms, governments can choose to change the incentives facing automotive manufacturers, or change the incentives for consumers, or both. We thus initially divide policies according to whether they target the incentives for auto manufacturers ("manufacturer" policies), or target individuals or organizations that purchase NGVs ("consumer" policies).

A further distinction can be drawn between policies that use financial incentives, such as subsidies, loans, or taxes ("economic" policies), and non-financial incentives ("regulatory" policies), such as technology and performance standards, and road access rights (IEA, 2002). Even without explicitly addressing consumer education these policies can be expected to have large distributional effects (Hughes and Urpelainen, 2015). The categorization of policy instruments is summarized in Table 2, along with representative policies identified in each category.

Using this framework we identify whether a policy is applied in each category, focusing on PHEV, BEV, and FCEVs. A key goal of the paper is to identify whether policies are technology neutral. We code a policy instrument as neutral if each NGV technology is addressed through a given policy intervention. We utilized a number of different sources of data in preparing the empirical section. While many countries have publicised vehicle electrification roadmaps, summarizing national policies, this is not the case in all instances. Policies targeting manufacturers, in particular, are not recorded in a single document. For each country case we followed a standard procedure. We first identify policy targets announced by national governments. We then identified primary document sources, in the form of government policy statements or budgetary information, and documentation by auto manufacturers. We rely on data from research institutes and similar sources where necessary. In addition, we asked national experts to review the case summaries to ensure information is valid and exhaustive.

Three limitations of the data are also noted. First, we focus on policies implemented by national governments, and not sub-national governments. The greater variety of policies, and differences in the level to which transport and fiscal policies are devolved to sub-national governments, mean summarizing policies at both levels of government is too complex a task to carry out in a single

Table 2
Categorization of NGV Policy Instruments, with examples.

Recipient\Instrument	Economic	Regulatory
Manufacturers	Subsidies for research and development, or to support manufacturing	Production or deployment targets for NGVs + regulatory compliance rules (e.g. safety, interoperability standards)
Consumers	Purchase subsidies & tax incentives for NGV consumers	EV license plates, preferential lane access for NGV vehicles, parking privileges

article. Second, our definition of technology neutrality does not include policy support for alternative fuels – primarily biofuels – in the analysis. This is reasonable in that concerns about lock-in are significantly driven by the network externalities of different technologies, due to requirements for different refuelling infrastructure. Omitting biofuels from the analysis, on the other hand, means that the technology neutrality of policies is assessed in a narrow sense; a broader analysis could include biofuels or synthetic fuels, and indeed policy support for advanced internal combustion engines or autonomous vehicles technologies, in assessing government approaches to de-carbonization in the transport sector. Finally, while maritime transport, air transport, and heavy road vehicles are also important contributors to GHG emissions in the transport sector, we follow Van den Steen et.al (2015) in limiting the analysis to passenger vehicles.

In the next section, we offer a qualitative summary of policies for each case, including introducing relevant EU legislation. We then summarize this data at the end of the section. Cases are introduced in descending order of market size, measured by the latest annual NGV passenger car sales. We first discuss national targets, and then introduce specific policy instruments.

4. Results

4.1. China

4.1.1. Targets

The Chinese central government adopted deployment targets for NGVs. A deployment target for New Energy Vehicles (NEVs) – defined as hybrid electric vehicles (HEVs), PHEVs, BEVs and FCEVs (Gong et al. 2011) – announced in January 2009 included a sales target for passenger vehicles of five percent between 2009 and 2011 (**Wang and Kimble, 2011:318**; Gong et.al. 2011). An additional deployment target was announced by the State Council as part of the 2009 "Ten Cities, Thousand Vehicles" program, with a sales goal of ten percent by 2012, 500,000 and five million of electric vehicles by 2015 and 2020, respectively. Wan et al. (2015, 116) note FCEVs were excluded from this deployment target (State Council, 2012).

China's fuel efficiency standards provided an additional incentive to manufacturers to produce NGVs. The government established a Corporate Average Fuel Consumption (CAFC) standard of 7 l/100 km (equivalent to 167 g of CO2/km) by 2015 in Phase III (2012-2015) of the national fuel efficiency standards (GB 27999–2011). In 2012, the State Council issued "China's Energy-Saving and New Energy Vehicle Industry Development Plan" (2012–2020) targeting a fleet-average fuel consumption of no more than 5 l/100 km (56mpg) in 2020. The CAFC standard was technology neutral in a broad sense, in that it does not mandate particular technologies, but provides an incentive to improve manufacturer fleet efficiency.

4.1.2. Policy instruments

As with other governments, early government support for NGVs began with public investment in research and development (R&D). NGVs were included in China's National High Technology R&D Program for 2001–2005 targeting FCEVs, BEVs, and PHEVs. The State Council designated NGVs as one of seven "Strategic and Emerging Industries," in October 2010 (Howell et al., 2014).

A range of financial incentives was provided by the central government to manufacturers to support R&D. The government provided loans and grants of 4.7 billion yuan (roughly USD 0.7bn), in the 11th Five Year Plan (2006–2010), with larger amounts invested by subnational governments (Howell et al., 2014; Gong et al., 2013, 212). On the consumer side, the 2009 "Ten Cities with Thousand Vehicles" program subsidized purchases by public sector organizations, including taxis, buses, and sanitation vehicles, with transfers made from the centre to local governments. The program also supported private purchases of NGVs in five cities focusing on PHEVs and BEVs only, with FCEVs excluded (Gong et al., 2013, 223). Nevertheless, FCEVs receive a buyer subsidy in 2016 and 2017-18 (ICCT, 2017, 6).

The government revised these incentives since their introduction in 2009. In 2013 subsidies were extended until 2015 and the "Ten Cities" program was refocused to a few pilot regions. Under fuel efficiency standards, ICEs and conventional hybrid vehicles also qualified for subsidies if the vehicle fuel economy outperformed the Phase III standard value (ICCT, 2013). The subsidy rate was extended and phased reductions were announced with the goal of eliminating subsidies by 2020 (ICCT, 2017; Wolter and Scherf, 2016). In total, the government spent 33.4 billion yuan (approximately USD 5 billion) on subsidies in 2015, with 330,000 NEVs produced, and more than 207,000 BEVs and PHEVs registered in 2015 (Ministry of Finance, 2016; ICCT, 2017; IEA, 2016).

The central government has thus used financial policy instruments targeting both manufacturers and consumers. It has used regulatory policy instruments to promote NGV development and deployment. To limit misconduct, the Chinese Government initiated an approval process for NEV models (Reuters, 2017). On-board monitoring devices became a subsidy requirement to verify the production, sale, and use of NGVs, with penalties introduced to enforce the anti-fraud policy (ICCT, 2017). Regulatory measures also targeted consumers: NGVs were made exempt from road tolls, the license plate lottery and value-added tax.

Taken together, the central Chinese government has used financial and regulatory policy instruments targeting both manufacturers and consumers. Further, the approach has been largely technology neutral in the narrow sense, although an important exception is the exclusion of consumer subsidies for FCEVs under the "Ten Cities" program, which has been central to the deployment strategy adopted by the government.

4.2. United States (US)

4.2.1. Targets

While the U.S. federal government has supported advanced vehicle technologies development since at least the 1970s, recent

efforts were underpinned by a deployment target of "1 million advanced technology vehicles on the road by 2015" announced by the Obama Administration in 2008 (USG, 2008). Like China, NGVs were thus supported by a deployment target.

Targets incorporated into Corporate Automotive Fuel Efficiency (CAFE) standards provided an additional incentive to producers to develop NGVs (Al-Alawi and Bradley, 2014). The 2007 Energy Independence and Security Act (EISA) established a mandated target for manufacturer fleets of passenger cars, light trucks, and medium-duty passenger vehicles regardless of technology type. New standards were issued after remaining unchanged for almost twenty years and were raised again in 2012 for the years 2017–2025 (USEPA, 2011). These are technology neutral, in that they mandate fleet-wide emissions improvements, including through improving traditional ICE efficiency.

4.2.2. Policy instruments

The 2008 deployment target was supported by financial incentives. The EISA provided subsidies to support manufacturers in improving vehicle efficiency. A US$25 billion federal loan targeting manufacturers under Section 136 of the Act, later expanded to include component makers under the title of the Advanced Technology Vehicles Manufacturing (ATVM) (United States Government Accountability Office, 2001, 1ff.). The funding legislation defined advanced technology vehicles as "ultra-efficient vehicles" with a fuel economy of at least 75 miles per gallon (3.8 l/100 km), among other conditions (Canis and Yacobucci, 2015, 9). The loan program was thus technology neutral, as vehicles meeting efficiency targets were deemed eligible. By 2011, 62 percent of the loans had been invested in ICE technologies (USGAO, 2011, 13). Financing under the AVTM did, however, require vehicles or components to be manufactured or designed within the United States, and the United States Department of Energy (DoE) used local employment to assess projects.

The DOE's Vehicles Technologies Office also financed the research, development, and testing of advanced vehicle technologies, including batteries, electric drivetrain, combustion, materials, and fuels (USDOE, 2016a). The funding also applied to stakeholder support and education. In FY2016, its budget was US$310 million, supporting research into lithium battery technologies, lightweight materials, and emissions control in ICEs (USDOE, 2016b). The DOE programs thus targeted a range of technologies, with a central focus on the provision of financial support to manufacturers.

The federal government also addressed consumer incentives. The 2008 Energy Improvement and Extension Act, amended by the 2009 American Recovery and Reinvestment Act, provided a tax credit of up to US$7500 to consumers purchasing BEVs, with the credit phasing out once sales reached 200,000 units from 1 January 2010. An equivalent tax credit was available through the Fuel Cell Motor Vehicle Tax Credit for up to US$8000 for fuel costs. An Alternative Fuel Excise Tax Credit also enabled organizations to take a credit for fuel costs from compressed natural gas, liquefied natural gas, liquefied hydrogen, propane, P-Series fuel, liquid fuel derived from coal, and compressed or liquefied gas derived from biomass. While specific tax credits targeted individual technologies, collectively these measures are coded as technology neutral.

Procurement policies also targeted demand. The Department of Defense was required to prioritize electric or hybrid technologies under U.S. Code 2922 g. Similar requirements existed for the federal car fleet for hybrid electric vehicles, fuel cell vehicles, and advanced lean burn vehicles, under the EISA and Executive Order 13693, issued in March 2015. The federal government also provided US$55 million to 2020 to support the deployment of low and zero-emission public transport from local, state, the federal governments, and non-government-entities through the Fixing America's Surface Transportation Act of 2008. Subnational bodies also received funding under the program to purchase electric buses and charging infrastructure (USFTA, 2015). We also code this as technology neutral, as the terms of the funding did not differentiate between NGV technologies.

Finally, the US Department of Transport administers High Occupancy Vehicle (HOV) lane access under 23 US Code 166, and allows local authorities to grant HOV privileged access to alternative fuel, and low emission and energy efficient vehicles. While subnational governments have a role to play in the implementation of HOV access rights, the US federal government enables this through the use of regulatory benefits targeting consumers. Furthermore, the relevant law enables privileged treatment to vehicles defined as low emission and energy efficient, and is thus technology neutral.

In summary, like China, the United States thus used a full range of both financial and regulatory incentives, targeting both manufacturers and consumers, in order to promote NGVs. The contrast with China is that consumer incentives remained targeted across all NGV technologies, and indeed often expanded to be technology neutral in a broader sense by incorporating biofuels and other technologies.

4.3. Japan (J)

4.3.1. Targets

The Japanese Government formulated is next-generation vehicle policy priorities in the 'NGV 2010'. The strategy set deployment targets across NGV technology types, with a FCEV target of 1 percent of vehicles on the road by 2020, and 3 percent by 2030. For BEV and PHEVs, the target was 15–20 percent of vehicles on the road by 2020, and 20–30 percent by 2030.

An additional deployment target was announced in 2015, envisioning that NGVs should represent five to seven percent of new vehicle sales by 2030 (GOJ, 2015a). The inclusion of BEVs, PHEVs, and FCEVs shows the government adopted a technology-neutral approach to NGV deployment.

Japanese fuel efficiency standards also provide an incentive to manufacturers to develop and deploy NGV technologies. Standards were implemented under the technology neutral "Top Runner" program, mandated under the Energy Conservation Law. Both domestic manufacturers and importers of more than 2000 vehicles are covered by the standards. Passenger cars were added to the Top Runner standards in 2007, targeting vehicle efficiency improvements using a sliding scale depending on vehicle weight.

4.3.2. Policy instruments

The central government provides R&D subsidies to manufacturers, targeting BEV and FCEV, as well as diesel, technologies. The largest sum was allocated to BEVs, with 5.6 billion Yen (US$50 million) provided for battery development, against 4 billion Yen (US $36 million) for FCEVs, and 500 million Yen (US$4.5 million) for diesel technologies.

A number of additional policies were implemented at the sub-national level through the "EV/PHV Town" initiative. Regions were selected, beginning March 2009 - with the second round of applications in December 2010 – with the national government then helping develop a series of best practices for promoting NGVs, included measures such as information provision and labelling, and the deployment of NGVs in local public transport. Although not included in the title of the program, FCEVs were not excluded from participating, and prefectural initiatives such as those implemented by Kanagawa Prefecture included FCEVs. We thus include these measures because of central government support and code them as technology neutral.

On the consumer side, both tax and subsidy instruments were applied. The Ministry of Land, Infrastructure, and Transport oversees an "Eco-car Tax Reduction", and "Preferential Measures on Greening Autos", applied to vehicles that outperform established emissions or fuel consumption standards, meaning taxes were technology-neutral. In addition, under the 2014 Clean Energy Vehicle subsidy, a fixed payment by vehicle technology was made to consumers in order to lower the purchase price of a vehicle and targets BEVs, PHEVs, and green diesel vehicles (GOJ, 2015b, 11, 12f). Under the listed vehicles no FCEVs were included (NGVPC, 2015). By 2017 Toyota's MIRAI and Honda's Clarity FCEV were included. The policy setting can thus be coded as technology neutral.

Regulatory measures targeting consumers are introduced at the prefectural and local-government levels through measures such as preferential parking treatment in urban centres. Administration of the measures reflect, once again the particular mix of central and local government partnerships introduced under the "EV/PHV Town" program, which includes a mix of financial and regulatory measures, and offers considerable independence for local authorities in developing and implementing measures.

Taken together, the Japanese national government thus adopted a similar mix of policies to those in other countries, using financial instruments targeting both manufacturers and consumers to promote research, development, and deployment of BEVs, PHEVs, and FCEVs, and including regulatory benefits for consumers, in addition to deployment targets for manufacturers.

4.4. European Union (EU)

4.4.1. EU policy instruments and targets

The EU provides a framework for EU member state NGV policies and is thus briefly described here. The EU issued three directives in 2009. The Renewable Energy Directive (2009/28/EG) instructed member states to cover 10 percent of their transport energy needs by renewable energies by 2020 as part of the EU 2020 Climate and Energy Package. EC Directive 443/2009 then required automobile producers to reduce fleet average emissions of new vehicles below 130 g of CO2/km by 2015 and 95 g/km by 2020. Finally, the European Commission adopted the Low Carbon Fuel Standard (LCFS), an amendment (2009/30/EG) to the 1998 Fuel Quality Directive (1998/70/EC), targeting a reduction of life-cycle emissions of 10 percent per unit of energy between 2010 and 2020. The Fuel Quality Directive took a well-to-wheel approach, containing specifications under article 7a reducing the greenhouse gas intensity of energy for road vehicles. The Directive (2009/30/EG) considered the use of biofuels, the replacement of gasoline and diesel with natural gas, and the use of clean electricity in EVs, thus adopting a technology-neutral approach.

The EU also provided R&D subsidies for manufacturers. The 2008 European Fuel Cells and Hydrogen Joint Undertaking (FCH JU) launched under the 6th European Framework Program for Research (FP6, 2008–2013) has to be mentioned. The partnership between the European Commission, industry and research organizations was established to accelerate the development and deployment of hydrogen and fuel cell technology (AFHYPAC, 2014). The European Council continued the Fuel Cell and Hydrogen Joint Undertaking (FCH-JU 2) under the Horizon 2020 framework (2014–2020) with a budget of 1.33 billion Euro, half funded by the European Commission (SETIS magazine, 2015). The EU is also a funding partner for member countries. Since 2013, the "EV Radar" of the European Commission's Joint Research Centre (JRC) provides an overview of EU and member country (co-) funded EV (PHEV & BEV) Research, Development, and Demonstration projects with a total budget of more than 1 million Euro each (EU Science Hub, 2016). Currently, the radar counts 230 projects in 16 EU member and 2 EFTA countries of more than 570 million Euro. Taken together, EU directives provide a technology neutral framework in combination with R&D subsidies, while enabling individual member states to implement EU-level policies.

4.5. Germany

4.5.1. Policy targets

In 2009, the German Federal Government adopted the "National Development Plan for Electric Mobility" (NEPE), including a target of one million electric vehicles deployed by 2020. The target does not distinguish between plug-in electric (PHEVs or BEVs), and hydrogen fuel cell electric vehicles (BMU, 2009). The NEPE aims to turn the German auto industry into a lead supplier, and the country into a lead market, for electric mobility.

The government also adopted the "German Federal Energy Concept" at the end of 2010, according to which the entire transport sector is required to increase efficiency by 10 percent by 2020, and 40 percent by 2050 (Schafhausen, 2012, 16). In addition, the "Government Program for Electric Mobility" of 2011 announced a target of 16 different EV models of German brands for 2014, at least 10 percent ultra-low carbon vehicles (ULCVs) in the Government vehicle fleet by 2015, and a total of six million NGV deployed by 2030 (BR, 2012). Policy objectives are thus fully technology neutral.

4.5.2. Policy instruments

The automotive industry and the German government started technology co-development of FCEVs through the "National Hydrogen and Fuel Cell Technology Innovation Program" (NIP), established in 2006 (BMU, 2007). With a total funding volume of € 2.1bn (50% government, 50% industry) for 2008–2015, and a second funding cycle beginning in 2016, the program focuses on financial support for the FCEV production and deployment ecosystem.

The "Integrated Energy and Climate Program" (IEKP) of 2007, which translates the EU 20-20-20 by 2020 goals into German policy, focuses on the role BEVs could play in managing renewable energy intermittency. Backed by the Economic Stimulus Package II (KoPa II), €500 million was provided for FY2009-11 to prepare the German market for the full spectrum of NGVs through the "Electric Mobility in Pilot Regions" demonstration program (BMVBS, 2012). It was extended with modifications for 2012-15.

The government also announced a doubling of expenditures for PHEV & BEV-related R&D (2009–2013) through the 2011 "Government Program for Electric Mobility," tied to the "German High-tech Strategy," and linked to R&D in battery and materials technologies and the role of Information and Communication Technology (ICT), in so-called lighthouse projects (2012–15). Under the NIP scheme, the Clean Energy Partnership (CEP) was designated a lighthouse project dedicated to fuel cell powered vehicle fleets (buses and cars) and the construction of 50 hydrogen refuelling points by 2015. Industry contributed €17 billion for R&D on alternative drivetrains (PHEVs, BEVs, and FCEVs) as part of these lighthouse projects, with €10-12 billions from the automotive industry (BR, 2012).

In 2011, financial policies were also introduced in support of NGV deployment targeting consumers. Tax cuts were introduced for electric company cars, along with a 10-year waiver of the motor vehicles tax. In April 2016, a buyer's premium of €4000 per BEV and €3000 per PHEV was introduced, limited to vehicles below €60,000 basic price. FCEVs were not considered, however, the 10-year waiver of motor vehicle tax at first registration introduced in 2012 was extended to 2016 and beyond, and applied to FCEVs as well.

The German Federal government also applied regulatory measures targeting consumers. In 2015, it introduced an "Electric Mobility Law" (EmoG) regulating the labelling of PHEVs and BEVs, and enabling municipalities to give exclusive access to bus lanes and free parking.

In sum, the German NGV policy was largely technology neutral. All three NGV technologies received financial support targeting producers. This was also the case on the demand side, although the type and magnitude of support varied across technologies.

4.6. France

4.6.1. Policy targets

The French Government adopted ambitious targets for NGV deployment. By 2015, newly registered vehicles were proposed to reach 7 percent electric propulsion, equivalent to 450,000 vehicles, 16 percent in 2020 (2 million EVs), and 27 percent in 2025 (4.5 million) (Leurent and Windisch, 2011). The Government further specified that half of newly purchased government vehicles should be plug-in electric by 2016. The target of full electrification of the country's bus fleet and other public service vehicles by 2025 includes FCEVs (Inside EVs, 2014; MTES, 2015; MTES, 2016). The French government thus adopted deployment targets that were technology neutral.

4.6.2. Policy instruments

In 2007 the French state declared the beginning of the (re-) electrification of vehicle propulsion through the "Grenelle Environnement," an annual public policy roundtable convening all national stakeholders. The Grenelle Environmental Forum translated EU Directive EC 443/2009's average fleet emissions standards into French national law.

In the fourth funding cycle for France's land transport research program (2008-12), the French Government spent €200 million (USD224 million) on decarbonized vehicle technologies across a number of research centres (TI, 2011). In response, Renault and PSA/Citroen forged an alliance with Japanese BEV manufacturers Nissan and Mitsubishi (Schamp, 2014, 6). A purchase agreement among 20 industry partners guaranteed the two leading French auto manufacturers vehicle demand of 50,000 EVs over the course of 5 years. La Poste, for example, ordered 15,600 fully electric Kangoo ZE at the end of 2011 (Le Figaro, 2011).

In addition, the 2009 "Automobile Pact" provided manufacturers with €6.5 billion in low-interest loans over five years, with a share allocated to clean vehicle innovation. Investments made into "vehicles of the future" under the "Grand Emprunt" provided another €750 million for R&D projects under the French Environment and Energy Agency (ADEME) for both BEVs and PHEVs such as the French-German EV test project CROME (ADEME, 2011). Also ADEME's EV demonstration fund was augmented by €50 million to a total of €80 million, and additional funding was made available for a battery manufacturing site in France and research that examines battery ecology and recycling (PR, 2009; TI, 2011; Leurent and Windisch, 2011).

In 2009, the National Research Agency (ANR) launched the "Hydrogène et piles à combustible" (HPAC) network specializing in hydrogen production through water electrolysis, hydrogen storage, and applications in the fuel cell (2009–11) (Leduc et al., 2010). The development of hydrogen fuel cell technologies was made legally manifest in article 37 and 121 of the "Loi sur la transition énergétique", the French Energy Transition Act of 2015. Like Germany, although the specific instruments and amounts differed, manufacturer support targeted both BEVs and FCEVs, and was thus coded as technology neutral.

A substantial part of the Government program is dedicated to a sophisticated consumer subsidy. The "bonus-malus" system of 2008 supports consumers purchasing vehicles with lower CO2 emission values than the annual target, and penalizes them for vehicles with higher CO2 emissions values. Long before, the French Government introduced a sticker called 'green vignette' that would be placed visibly on the front window of the car to recognize cleaner road vehicles (Légifrance, 1999). The French Government also introduced a diesel scrappage program in 2009 that provided a subsidy for BEVs and PHEVs. A 2016 revision of the "bonus

Table 3
Presence of Policy Instrument by Country.

Recipient\Instrument	Economic	Regulatory
Manufacturers	CH, US, JP, DE, FR	CH, US, JP, DE, FR
Consumers	CH, US, JP, DE, FR	CH, US, JP, DE

Note: CH = China; US = United States; JP = Japan; DE = Germany; FR = France. EU-level policies that have mandates for member states are included in coding for EU member countries (DR and FR).

écologique" gave €6500 to consumers for a PHEV and €10,000 for BEVs when replacing a diesel vehicle of 15-years or more by a new vehicle of emissions below 60 g/km (MEDDE, 2015). Beginning with the fiscal year 2016/17, the French Government reduced the eligibility criteria to 10-year old diesel vehicles to phase out the fuel tax advantage diesel vehicle holders receive until 2020 (CHB44, 2015).

To conclude, the French Government has invested more than one billion Euro into supply-side vehicle electrification, while domestic manufacturers and other relevant actors also benefited from EU R&D funding. To meet EU legal provisions, the French Government developed an advanced system addressing vehicle CO2 emissions in a largely technology neutral way subsidizing more efficient ICEs, hybrid electric vehicles, and NGVs at purchase, although the type and magnitude of support varies across technologies.

5. Conclusions

In this article we developed and employed a framework for comparing policies employed in support of the development and deployment of next generation vehicles in leading automotive markets. We also examined whether national policies supporting NGVs are technology neutral. Automotive companies have developed and commercialized multiple technologies - centered on battery electric vehicles and fuel cell electric vehicles - in contributing to decarbonizing the transport sector. Governments face an important choice in determining whether, and for how long, to provide policy support for these competing technologies. On the one hand, technologically neutral policies decrease the likelihood that suboptimal technologies become locked-in as the dominant technology. On the other hand, it may not be optimal for governments to continue to support multiple competing technologies as a market matures and one technology begins to dominate.

Our results suggest that governments are using a remarkably similar portfolio of policies in supporting NGVs development and deployment, centring on R&D subsidies targeting manufacturers, consumer subsidies, and a mix of regulatory policies (see Table 3). This use of similar instruments is surprising, given important differences in the political institutions across the countries studied here, and different national systems of innovation.

An important caveat is that the levels of spending from each government differ, even as the portfolio of policies are similar in terms of focus (both producer and consumer), and instrument (both economic and regulatory). Aggregating total public spending across policies and countries is challenging given regulatory instruments, such as preferential lane access, are not readily comparative in fiscal terms. Nevertheless, it is plausible that while governments are employing a similar portfolio of policies, the balance of spending across the categories of policy instruments is different.

Our findings also show that although there are some exceptions (Table 4). In the German case, for example, the purchase premium introduced in 2016 did not apply to FCEVs, although this may be explainable by the later mass commercialization of FCEV models, relative to BEVs. Governments are maintaining technology neutral approaches despite differences in governing institutions, and also despite automotive companies in the different national jurisdictions making different choices about which technologies to develop and commercialize.

We further note that while policies are largely technology neutral, the different stages of industry maturity means their effect may favour BEVs over FCEVs. The distinction between technology neutrality in policy and neutrality in outcomes thus emerges as potentially important because of differences in the product cycle, and the presence of network externalities in NGVs. BEVs were commercialized earlier than FCEVs, and manufacturers have produced more models. It is plausible that the availability of consumer subsidies to support the diffusion of BEVs thus provides an advantage to this technology, regardless of the neutrality of policy

Table 4
Technology Neutrality by Country/Policy Instrument.

	Manufacturers			Consumers		
	PHEV	BEV	FCEV	PHEV	BEV	FCEV
China	√	√	(√)	√	√	√
United States	√	√	√	√	√	√
Japan	√	√	√	√	√	√
Germany	√	√	√	√	√	(√)
France	√	√	√	√	√	(√)

Note: () parentheses indicate at least one sub-policy does not include the indicated technology.

instruments used to support alternative propulsion systems, given the presence of network externalities. BEVs and FCEVs require charging infrastructure, and the products are more valuable in the presence of this infrastructure. Consumer subsidies thus provide a competitive advantage to PHEVs and BEVs in the presence of these network externalities, which may lead to technology lock-in even if consumer subsidies are capped numerically by technology, as is the case in the United States.

In the next section, we discuss possible explanations for these results and their broader implications for NGV deployment.

6. Discussion

How can we explain the similar portfolio of policies utilized by national governments in the countries examined in this study? Wesseling (2016) identifies government institutions, specifically the degree to which interests are coordinated within "Varieties of Capitalism," (Hall and Gingerich, 2009), and the influence of industrial interests, as two possible explanations for variation in patterns of support for NGVs. Our results do not provide strong support, however, for a causal relationship between the structure of governing institutions, and the types of policies adopted. Each of the countries we examined have a domestic automobile industry. The application of a broad range of policies is thus plausibly influenced by the interests of industry, although a causal interpretation would require examining additional country-cases that do not have an incumbent automotive industry. Process-tracing techniques could also be used to infer a causal relationship between the influence of industry groups and policy design and implementation (Beach and Pedersen, 2013).

More puzzling is the application of the principle of technology neutrality in the implementation of these policies. If industry groups are key actors determining policies, then it is reasonable to expect that technologies targeted for policy support are a function of the commercial choices being made by automotive groups. Yet although automakers are making different choices, with most focusing on PHEVs and BEVs, while a small number of automakers favour FCEVs, we see largely equal treatment of technologies across countries.

A plausible explanation for this outcome is that governments are balancing the interests of domestic automotive companies in increasing their competitiveness in new product markets, with the broader policy goal of ensuring the dominant technological design that emerges as the market matures maximizes consumer and environmental benefits. An additional plausible explanation worthy of investigation focuses on the role of interdependence (Meckling and Hughes, 2018). Cross-national policy learning facilitated through organizations such as the International Energy Agency could plausibly lead governments to adopt similar approaches, for example. It is also possible that Global Supply Chains, which play an important role in structuring the production in the automotive industry, lead firms to support the harmonization of policy support across national markets. Careful examination of the processes of policy development across different national contexts offers one strategy for examining the validity of different explanations for the observed pattern of policies being implemented.

In addition to examining the explanatory power of theories that might account for outcomes, the findings in this article can be extended in at least three ways. First, in this article we have focused on policies implemented by national governments in support of NGVs. Sub-national governments also play an important role in hastening the deployment of NGVs and these policies are worthy of consideration. In addition, governments are supporting the deployment of charging infrastructure, which differs by technology and vehicle type (light versus heavy-duty). Further research should examine the comparative public policies at sub-national levels of government, and across charging infrastructure, discerning more closely between light and heavy-duty road vehicles.

Second, while we have focused on public policies in support of NGVs, spending by private companies also plays a crucial role in determining the competitiveness of different clean energy technologies (Binz et al., 2017). Legacy automakers are clearly making different choices about which technologies to support, as shown by Toyota's and Mercedes' support for FCEVs, and this is likely to influence which portfolio of technologies emerge as dominant. An examination of patterns of private sector investment in NGVs is thus warranted.

Finally, we have focused on BEV, FCEV, and hybrid technologies, however there is a broader range of options available, notably those that reduce the carbon content of fuels used in the dominant fleet of internal combustion engines. Incorporating policies targeting technologies such as biofuels is thus also warranted, in order to gain a full understanding of the types of policies governments are using in promoting decarbonization of the (road) transport sector, and the extent to which policies are technology neutral.

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